

Did congestion pricing make Manhattan taxis faster?

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Code: <https://github.com/palak348/FDE-Assignment/tree/main/assignment-2>

1. Problem and stakeholders

On 5 January 2025 New York began tolling vehicles entering Manhattan south of 60th Street (the Congestion Relief Zone) to reduce congestion. Before deciding whether to keep or adjust the toll, the MTA needs to know whether traffic in the zone actually moves faster. I also considered slow airport trips and suspicious fares, but airports have no queue data and fare anomalies have no clear decision behind them; the toll affects the whole city and allows a clean before-and-after test.

The MTA owns the toll and the KPI; the Taxi and Limousine Commission (TLC) owns the trip data. The KPI is the **median speed of taxi trips that start and end inside the zone, on weekdays between 5am and 9pm**, which are the peak toll hours. I compare January 2024 with January 2025.

2. Data sources and retrieval

I split the business question into five smaller questions, each matched to the information it needs and the system that owns it (Table 1). TLC trip records are the system of record. Retrieval uses files, JSON APIs and SQL (DuckDB); every download is checked for completeness and kept unchanged.

Table 1. Business questions mapped to the information and sources that answer them

Question	Information needed	Source (owner)	Grain	Retrieval and check
Did trips get faster?	times, distance, zones	Trip records (TLC)	one trip	Parquet; rows = file metadata
Which trips are tolled?	zone ID, toll boundary	TLC and MTA zones	one zone	CSV and API; 38 of 265 tolled
Was the toll charged?	fee on each trip	Trip records (TLC)	one trip	Parquet; 2024 fee = unknown
Could weather explain it?	daily rain and snow	Open-Meteo	one day	API; one record per day
Do official figures agree?	monthly zone speed	MTA	zone, month	API; count matches API

The main gap is that trip records have no GPS trace, so trips that only pass through the zone are invisible.

3. Data model and pipeline

The model centres on one event, the trip from pickup to dropoff, linked to its zone and its day, with the toll as the intervention that splits before from after (Figure 1, top). The pipeline (bottom) runs as one logged command; a month that fails a quality gate is held back, and reruns give the same output.

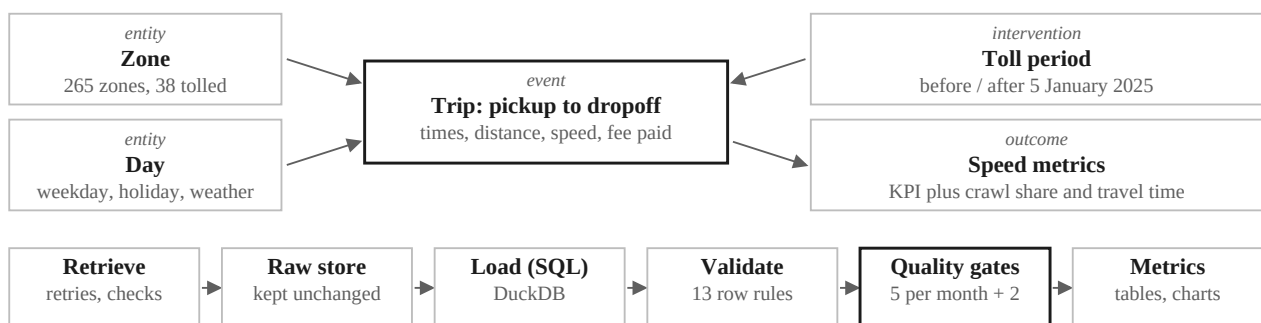
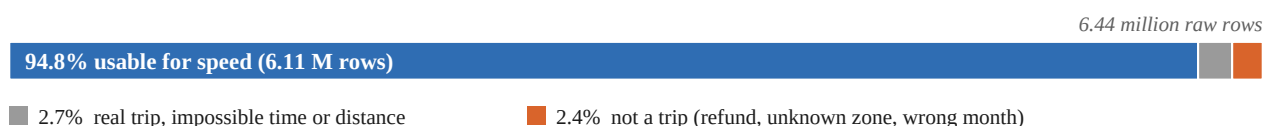


Figure 1. Workflow model (top) and pipeline (bottom).

4. Data quality

Thirteen business rules flag bad rows instead of deleting them, for example a trip shorter than one minute, an average speed above 60 mph, or a pickup outside the file's month. Profiling found a \$863,380 fare, a 312,722-mile trip, trips dated 2002, and a vendor whose trips all last zero seconds.



5. Results

On the raw figures every speed measure improved and zone demand grew 24% (Table 2).

Table 2. Weekdays 5-31 January, excluding public holidays. Speed metrics are based on 546,080 trips in 2024 and 673,564 in 2025.

Metric	Jan 2024	Jan 2025	Change
Median speed in the zone, 5am-9pm (KPI)	7.74 mph	7.89 mph	+1.9%
Trips crawling below 5 mph	12.8%	11.7%	-8.4%
Slowest 10% of trips (minutes per mile)	12.80	12.52	-2.2%
Taxi trips in the zone per weekday	38,419	47,652	+24.0%
Zone trips charged the toll	not applicable	98.4%	

6. Adjusting for weather

The raw increase of 1.9% cannot be taken at face value: January 2024 was far wetter than January 2025 (Figure 2), and bad weather slows traffic regardless of any toll. Both periods also start on 5 January, so the four pre-toll days of 2025 are not counted as "after".

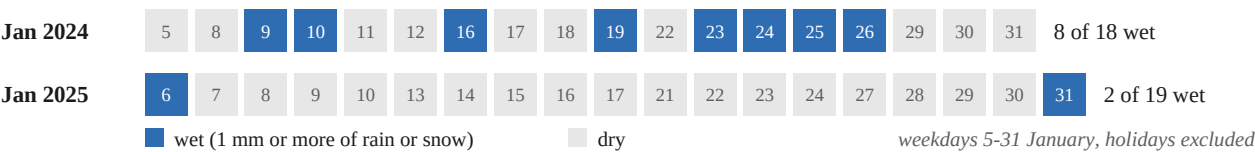


Figure 2. Wet and dry weekdays in each comparison window.

I therefore compared dry days only, inside the zone against outside it. Dry-day speeds rose 0.5% inside and fell 3.3% outside, a net improvement of about **3.8 percentage points**, in line with the MTA's own +3.8% for taxis and ride-hail cars in the zone (Figure 3, left). Gains run from late morning to evening; trips before 6am slowed slightly (right).

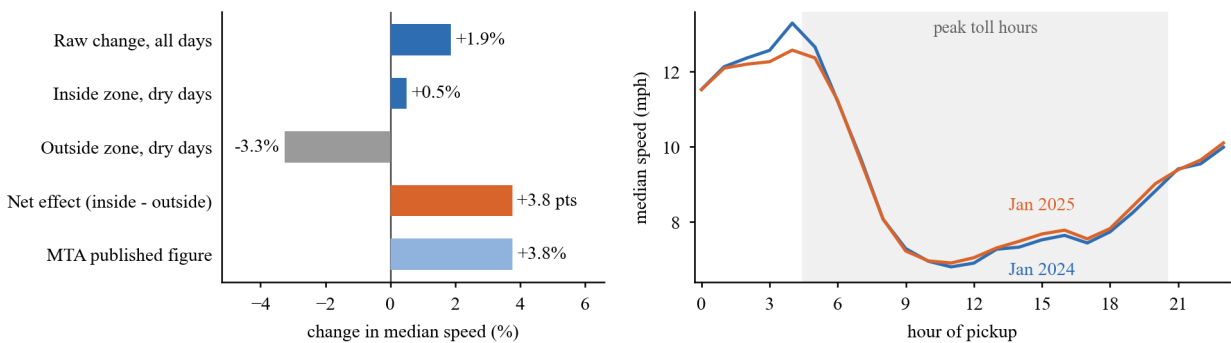


Figure 3. Left: raw change against the weather-adjusted comparison. Right: median speed by hour inside the zone on weekdays.

7. Facts, assumptions, unknowns and limitations

Type	Summary
Facts	Zone speed +1.9%, crawling trips -8.4%; +3.8 pts vs outside on dry days; toll charged on 98% of zone trips.
Assumptions	Whole-trip speed reflects zone traffic; negative totals are refunds; one weather point covers the city.
Unknowns	Pass-through trips (no GPS); why 1.6% of zone trips paid no toll; Uber and Lyft; whether the effect lasts.
Limitations	One month per side; 3.2 pts if trace snow counts as wet; control includes airports; not proof of cause.

8. Recommendation

The evidence supports keeping the toll: taxi trips in the zone became about 4% faster relative to the rest of the city, even though more taxi trips were made there. Because the result rests on a single month, the next step is to rerun the pipeline for February to June. I would also ask the TLC for a flag showing whether each trip entered the zone, the list of toll exemptions, and a fix for the vendor that records zero-second trips.